

K-means and K-means++: A Study on the Impact of Initialization in Clustering

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Abstract

This study investigates the impact of initialization strategies on the performance and stability of the k-means clustering algorithm. Two algorithms are implemented from scratch and compared: standard k-means with random initialization and k-means++ with distance-based probabilistic initialization. Both algorithms are evaluated on two datasets: a synthetic dataset with well-separated clusters and a subset of the MNIST handwritten digit dataset. Each algorithm is run for multiple trials and compared in terms of inertia and result variability. Results show that k-means++ produces significantly better and more stable results on the synthetic dataset, while the difference between the two algorithms diminishes on the high-dimensional MNIST dataset. These findings are consistent with the theoretical guarantees proposed by Arthur and Vassilvitskii (2007).

Keywords: K-means, K-means++, Clustering, Initialization, Unsupervised Learning

1 Introduction

K-means is one of the most popular unsupervised learning algorithm. The main goal of K-means algorithm is separating data points to k groups while the points in the same group should be as much as similar to each other. While the algorithm works, it works in 2 step loop. First, assign every point to the closest centroid, and then update every centroid as average of points which are assigned to that centroid. This loop continues until centroid positions don't change anymore. However, k-means has a well known disadvantage. Selection of starting centroids are really sensitive. Random chosen starting points can cause bad local minimum and algorithm can produce different results in every run. Arthur and Vassilvitskii (2007) suggested a new algorithm with better initialization to solve the problem: k-means++. Instead of choosing all starting centroids randomly, each new centroid is selected with a probability proportional to the squared distance from the existing centroids. This approach encourages starting centroids spread over data and leading to better and more consistent results In this project, both algorithms are implemented from scratch and evaluated on two different dataset. First one is the synthetic dataset with well seperated clusters and subset of MNIST handwritten numbers dataset. Each algorithm is run with multiple trials and compared the results of inertia changes.

2 Dataset

In this project, 2 different dataset is used.

2.1 Synthetic Dataset

The synthetic dataset is created by using numpy library. Data points are sampled with Gaussian distribution among 3 centers. Centers are determined as (1,1), (5,5) and (9,1), and standard deviation is set to 0.8. In total from each center 300 points are created and obtained 2 dimension dataset with total 900 points. These parameters were chosen deliberately. Because the dataset is 2 dimension, it leads us to visualize it on a scatter plot and can observe the behaviour of algorithm. With 0.8 standard deviation, clusters are separated easily from each other. This dataset is used to test the fundamental behaviour of algorithms and justify results with plots.

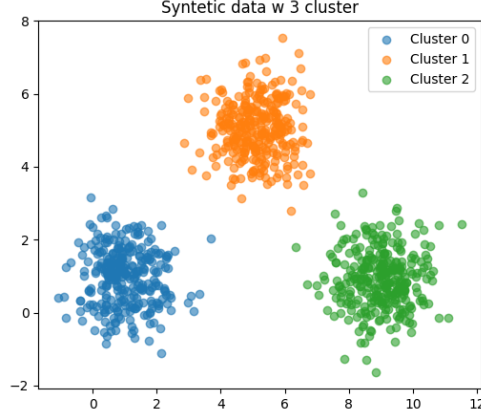


Fig. 1: Synthetic dataset — 3 clusters with 300 points each

2.2 MNIST Dataset

MNIST includes 70.000 pictures of handwritten numbers from 0 to 9. Each picture has 28x28 pixel, and turned to 784 dimension vectors. The dataset is imported via sklearn library. Since clustering is an unsupervised task, labels were not used during training and were only kept for reference. Because of the number of data in dataset, running a trial with 70000 point was hard to handle. That's why, from every number class, 500 sample is chosen randomly and created the sub-dataset with total of 5000 points. This approach ensures equal representation of every class while keeping computation time reasonable

3 Algorithms

3.1 K-means

K-means algorithm has 4 fundamental steps. First, k data points are randomly selected from data points and these are assigned as starting centroids. In the second step, every data point assigned to closest centroids. In this step, to understand and measure distances to choose closest centroids, Euclidean distance is computed. The formulation of Euclidean distance is:

$$d(a, b) = \sqrt{\sum_{i=1}^n (a_i - b_i)^2}$$

In third step, with taking average of points which are assigned to every cluster, centroids are updated. As a last step, second and third step are repeated until location of centroids no longer change anymore. For an approximation criteria, allclose function from numpy is used.

Inertia is the sum of the squared distances of each point from its centroid:

$$I = \sum_{i=1}^n \|x_i - c_{\ell(i)}\|^2$$

3.2 K-means++

K-means++ algorithm is so similar to original K-means algorithm. It just has different approach in initialization step, but all the rest is same. First centroid starting points are chosen randomly. For each subsequent centroid, the minimum squared distance from each point to the nearest already-selected centroid is computed. A probability distribution proportional to this distance is created, and the next centroid is sampled from this distribution. This process continues until k centroids are selected. With this approach, starting centroids are spread to data and decrease the probability of having bad local minimums. Arthur and Vassilvitskii (2007) theoretically demonstrated that this initialization strategy restricts the expected inertia to $O(\log k)$ times the optimal solution.

4 Experiments

In the experiment stage, both algorithms are tested on the synthetic dataset first and on MNIST second. While testing on synthetic dataset, k is selected as 3, because the dataset is created for exactly 3 clusters. Every algorithm is run 300 times. This number is enough to see impact of random initialization statistically meaningful and also appropriate choice for computation time. For the dataset MNIST, k is selected as 10 because of the number of different classes in dataset. Every algorithm is run 100 times. Due to the high dimensionality of the MNIST dataset, computation time is far longer than synthetic data for every trial, that's why number of trial for MNIST dataset is selected as 100. In every trial, inertia value is recorded. The results are compared on inertia, standard deviation and distribution graphics. The experiment is run on fixed random seed.

5 Results

5.1 Results of Synthetic Data

As a result of 300 trials, there is a significant difference in results is observed between k-means and k-means++ algorithms. K-means average inertia is 1474.96 and standard deviation is 1278.36. K-means++ has average 1114.3 inertia score and 281.41 standard deviation.

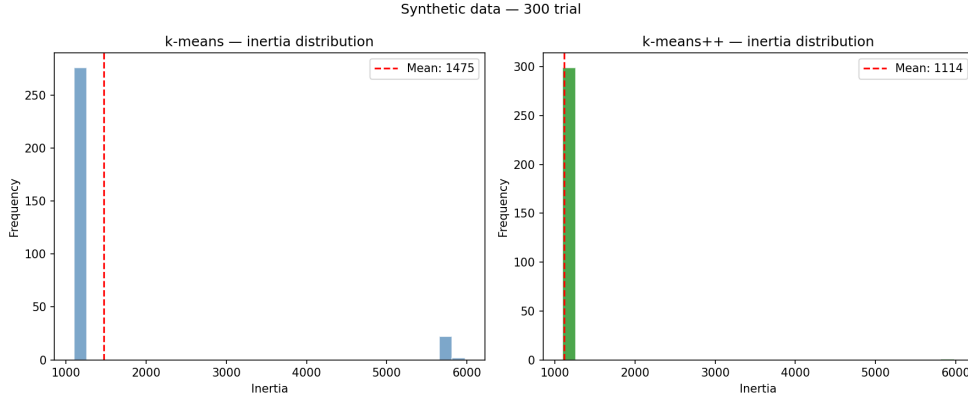


Fig. 2: Synthetic dataset — inertia distribution over 300 trials

When we check the histogram, even though k-means has mostly low inertia value, some trials has inertia values like 5980. These high values exist because of bad starting point of algorithm. However, K-means++ inertia values are in narrow interval for all the trials.

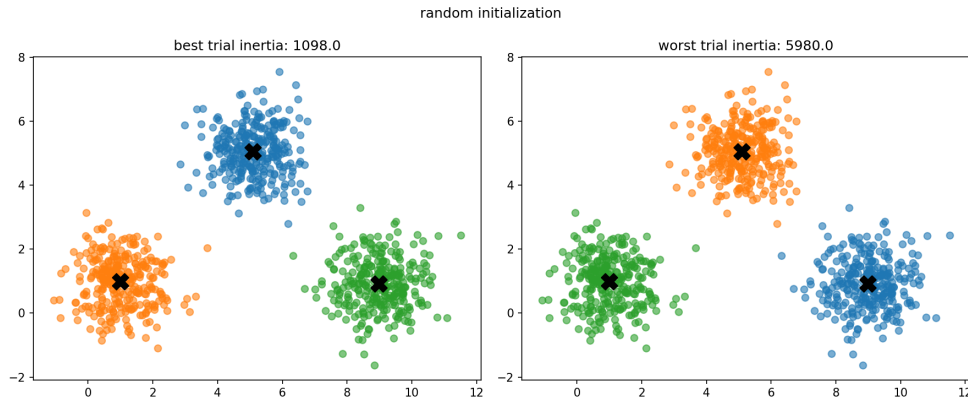


Fig. 3: Best trial (inertia: 1098.02) vs worst trial (inertia: 5980.38) for k-means

A comparison of the best and worst trials clearly demonstrates this. In the best trial (inertia: 1098.02), the three clusters are clearly separated, and the centroids are positioned exactly in the middle of each group. In the worst trial (inertia: 5980.38), two clusters merge, and the algorithm converges to an incorrect solution.

5.2 Results of MNIST Dataset

After 100 trials, the k-means mean inertia was measured as 12,842,467,321, with a standard deviation of 27,766,622. The k-means++ mean inertia was measured as 12,845,567,184, with a standard deviation of 37,330,196.

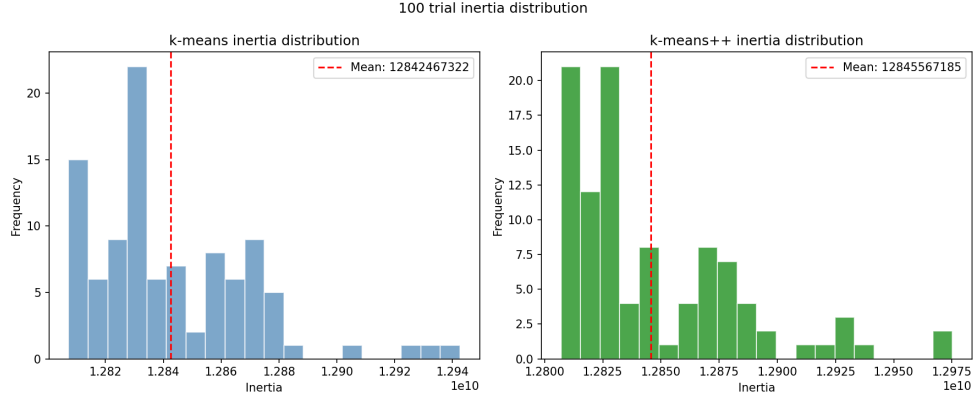


Fig. 4: MNIST dataset — inertia distribution over 100 trials

Unlike the synthetic dataset, the difference between the two algorithms is very small on MNIST. Interestingly, k-means achieved a slightly lower mean inertia than k-means++ in this case. This can be attributed to the high dimensionality and overlapping cluster structure of MNIST, where the advantage of smarter initialization becomes less significant.

5.3 Comparison of Both Datasets

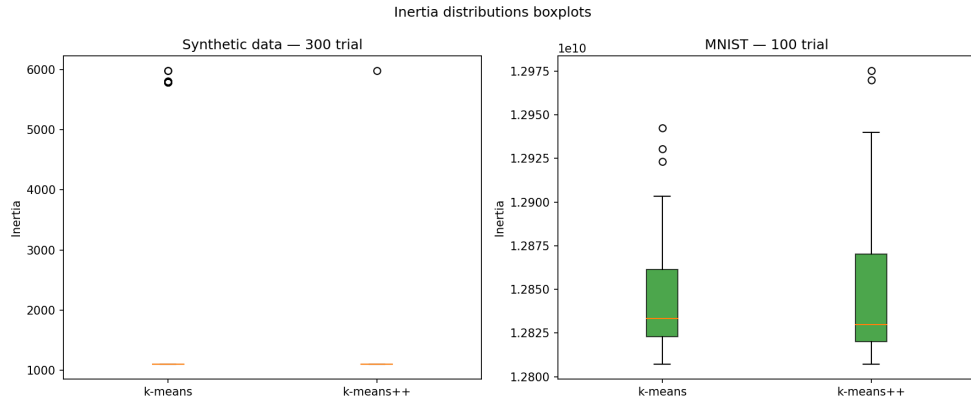


Fig. 5: Inertia distribution comparison — synthetic vs MNIST

The boxplot comparison summarizes the results for both datasets. On the synthetic dataset, k-means produces clear outlier trials with very high inertia, while k-means++ remains stable. On MNIST, both algorithms produce very similar distributions. This suggests that the advantage of k-means++ is most pronounced when clusters are well-separated, and diminishes in high-dimensional spaces with naturally overlapping clusters.

6 Conclusion

In this study, both k-means and k-means++ algorithms are implemented from scratch and compared on two different dataset. In the result of synthetic dataset we see that k-means++ produces significantly better and more accurate results compared to random initialization. While K-means struggles with bad local minimums in some trials, k-means++ stays in a narrow interval in all trials. This finding matches with theoretical guarantee of Arthur and Vassilvitskii (2007). In MNIST dataset, we observed a different situation. In this dataset with a high-dimensional and inherently overlapping cluster structure, the difference between the two algorithms has been significantly reduced. This demonstrates that the advantage of k-means++ depends on the data structure. This project has some limitations. For MNIST, a subset of 5,000 points was used instead of the full 70,000-point dataset due to computation time constraints. Also, only the Euclidean distance metric was used; different distance metrics may produce different results.

Declaration

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